

# 1.9.3

EE25BTECH11014 - Bhoomika Lokesh

**Question:**  $AOBC$  is a rectangle whose three vertices are  $(0, -3)$   $(0, 0)$   $(4, 0)$ . The length of its diagonal is\_\_\_\_\_.

**Solution:** Given the points **A**, **O** and **B** :

| Point   | vector                                  |
|---------|-----------------------------------------|
| Point A | $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$  |
| Point O | $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$  |
| Point B | $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$ |

TABLE 0: Vectors of the points

Determining the Coordinates of Point C:

$$\mathbf{A} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \mathbf{B} = \begin{pmatrix} 0 \\ -3 \end{pmatrix} \quad (0.1)$$

Since **C** is opposite to **O** in the rectangle,

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (0.2)$$

$$\therefore \mathbf{C} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (0.3)$$

We know that the length of the diagonal vector is magnitude of the vector **C**.

$$\mathbf{C} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (0.4)$$

$$|\mathbf{C}| = \sqrt{\mathbf{C}^T \cdot \mathbf{C}} \quad (0.5)$$

$$\mathbf{C}^T = \begin{pmatrix} 4 & -3 \end{pmatrix} \quad (0.6)$$

$$\mathbf{C}^T \cdot \mathbf{C} = \begin{pmatrix} 4 & -3 \end{pmatrix} \begin{pmatrix} 4 \\ -3 \end{pmatrix} = 4^2 + (-3)^2 = 16 + 9 = 25 \quad (0.7)$$

$$|\mathbf{C}| = \sqrt{25} = 5 \quad (0.8)$$

Therefore the length of the diagonal is 5.

See Fig 0.1,

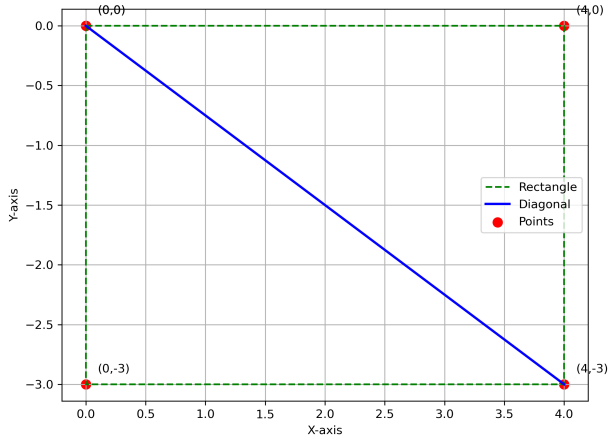


Fig. 0.1: Graph